

Notes 8

cat

Definition: The `cat` (concatenate) command is used to display, combine, and create files.

Usage / Formula:

```
cat [options] [file_name]
```

Examples:

```
cat file.txt                # Display contents of file
cat file1.txt file2.txt     # Concatenate two files
cat > newfile.txt           # Create a new file (overwrite)
cat >> file.txt              # Append to a file
```

tac

Definition: The `tac` command is the reverse of `cat`; it displays file contents in reverse order (last line first).

Usage / Formula:

```
tac [file_name]
```

Examples:

```
tac file.txt                # Display file in reverse line order
```

head

Definition: The `head` command displays the first few lines of a file (default is 10 lines).

Usage / Formula:

```
head [options] [file_name]
```

Examples:

```
head file.txt           # First 10 lines
head -n 5 file.txt      # First 5 lines
```

tail

Definition: The `tail` command displays the last few lines of a file (default is 10 lines).

Usage / Formula:

```
tail [options] [file_name]
```

Examples:

```
tail file.txt           # Last 10 lines
tail -n 20 file.txt     # Last 20 lines
tail -f logfile.txt     # Follow file updates in real time
```

cut

Definition: The `cut` command extracts specific columns or fields from text files.

Usage / Formula:

```
cut [options] [file_name]
```

Common Options:

- `-d` → delimiter
- `-f` → field number

Examples:

```
cut -d "," -f 1 file.csv # Extract first column from CSV
cut -c 1-5 file.txt      # Extract characters 1 to 5
```

sort

Definition: The `sort` command sorts lines in a file alphabetically or numerically.

Usage / Formula:

```
sort [options] [file_name]
```

Common Options:

- `-n` → numeric sort
- `-r` → reverse order

Examples:

```
sort file.txt           # Alphabetical sort
sort -n numbers.txt    # Numeric sort
sort -r file.txt       # Reverse sort
```

WC

Definition: The `wc` (word count) command counts lines, words, and characters in a file.

Usage / Formula:

```
wc [options] [file_name]
```

Common Options:

- `-l` → line count
- `-w` → word count
- `-c` → character count

Examples:

```
wc file.txt             # Lines, words, characters
wc -l file.txt          # Count lines only
wc -w file.txt          # Count words only
```
